

Final Project

Health Data Analytics with SAS

Divya Verma

BINF 5210 – Health Data Analytics with SAS

Department of Health Informatics

School of Health Professions

Rutgers University

dv441@rutgers.edu

Project Description



Background:

A study was conducted to determine whether a “DRUG A” helps to reduce the fasting sugar level. There are 3400 patients were enrolled for this study. They were randomly assigned to one of three treatment groups: Low-Drug A, High_Drug A, and Placebo. After ten weeks their reading was taken.



Research Question:

Is there a significant difference in mean fasting sugar levels (Test_Score) among the three treatment groups: Low Drug A, High Drug A, and Placebo after 10 weeks?



Hypothesis

Null Hypothesis (H0)

- $\mu_{Low} = \mu_{High} = \mu_{Placebo}$
- There is no difference in mean fasting sugar levels among the three groups.

Alternative Hypothesis (H1)

- H1= *Not all group means are equal.*
- At least one group has a mean fasting sugar level that is different from the others.



Appropriate Method to Evaluate the Research Question

Variables:

- The independent variable (Group) has 3 categories- High Dose, Low Dose and Placebo
- The dependent variable (Test_Score) is continuous

Test selection- based on assumptions-> Decide Parametric vs non-parametric test based on assumptions

Parametric test assumptions:

- 1) Normal distribution within each group (PROC UNIVARIATE)
- 2) Homogeneity of variance (Levene's Test through PROC GLM/ANOVA)

Recommended Test:

- If data is normally distributed, One-way ANOVA (because we are comparing means >2 groups. T-test is only for 2 groups.)
- If data is NOT normally distributed, perform a non-parametric alternative - Kruskal-Wallis Test

Prepare Data

(Raw Data: Two csv files)

First Study_Project1 = 3,233 rows

Variables	Datatype
Patient ID	Numeric
Age	Numeric
State	Character
Length of Stay	Numeric
Total Charge	Numeric

1	51 AZ	1	3170
2	50 AZ	2	4319
3	74 AZ	3	5850
4	59 AZ	1	6298
5	57 AZ	5	5386
6	71 AZ	4	4737
7	73 AZ	3	7525
8	49 AZ	1	2817
9	30 AZ	2	3430
11	20 AZ	2	3316
12	83 AZ	2	2922
13	45 AZ	3	5449
14	29 AZ	1	2493
15	72 AZ	3	5483
16	70 AZ	2	3988
17	30 AZ	1	3461
18	62 AZ	1	2865
19	30 AZ	2	2502
20	43 AZ	2	4566
21	81 AZ	5	4819
22	43 AZ	1	1717
23	52 AZ	2	1870
24	14 AZ	2	1710
25	77 AZ	2	1645

Second Study_Project1 = 3,200 rows

Variables	Datatype
Patient ID	Numeric
Group	Character
Test_Score	Numeric

1	Low	46.64957
2	High	35.35729
3	Placebo	87.62965
4	Low	44.59978
5	Low	59.04604
6	High	77.40581
7	Placebo	65.62726
8	High	49.46664
9	Low	42.71787
10	High	83.78794
11	Placebo	35.36306
12	Placebo	37.1483
13	Placebo	60.00429
14	Placebo	33.08712
15	n/a	41.17504
16	n/a	54.4942
17	Low	36.65837
18	High	99.3624
34	Placebo	48.30784
35	Low	69.02478
36	Low	51.70749
37	High	89.72073
38	Placebo	87.20592
39	High	54.41257
40	Low	79.97868

Step 1. Load Files

```
/* SAS PROJECT*/  
Libname L1 "C:\Users\dverm\OneDrive\MSHI COURSES\Current courses\SAS\SAS files";  
Filename Study1 "C:\Users\dverm\OneDrive\MSHI COURSES\Current courses\SAS\project\Datafiles\Study1.csv";  
Filename Study2 "C:\Users\dverm\OneDrive\MSHI COURSES\Current courses\SAS\project\Datafiles\Study2.csv";  
  
/* LOAD Study1*/  
Data L1.Study1;  
  Infile Study1 DSD DLM="," misover;  
  input PatientID Age State$ LOS Total_Charge;  
run;  
Proc print data= L1.Study1;  
  Title "Study1";  
run;  
/* LOAD Second_Study_Project1*/  
Data L1.Study2;  
  Infile Study2 DSD DLM="," misover;  
  input PatientID Group$ Test_Score;  
run;  
Proc print data= L1.Study2;  
  Title "Study2";  
run;
```

Study1

Obs	PatientID	Age	State	LOS	Total_Charge
1	1	51	AZ	1	3170
2	2	50	AZ	2	4319
3	3	74	AZ	3	5850
4	4	59	AZ	1	6298
5	5	57	AZ	5	5386
6	6	71	AZ	4	4737
7	7	73	AZ	3	7525
8	8	49	AZ	1	2817
9	9	30	AZ	2	3430
10	11	20	AZ	2	3316
11	12	83	AZ	2	2922
12	13	45	AZ	3	5449
13	14	29	AZ	1	2493
14	15	72	AZ	3	5483
15	16	70	AZ	2	3988
16	17	30	AZ	1	3461
17	18	62	AZ	1	2865
18	19	30	AZ	2	2502
19	20	43	AZ	2	4566
20	21	81	AZ	5	4819
21	22	43	AZ	1	1717
22	23	52	AZ	2	1870
23	24	14	AZ	2	1710
24	25	77	AZ	2	1645
25	26	42	AZ	2	3207
26	27	60	AZ	4	5017
27	29	65	AZ	5	7115
28	30	67	AZ	5	5198
29	31	61	AZ	4	8165
30	32	81	AZ	20	37133

Study2

Obs	PatientID	Group	Test_Score
1	1	Low	46.65
2	2	High	35.36
3	3	Placebo	87.63
4	4	Low	44.60
5	5	Low	59.05
6	6	High	77.41
7	7	Placebo	65.63
8	8	High	49.47
9	9	Low	42.72
10	10	High	83.79
11	11	Placebo	35.36
12	12	Placebo	37.15
13	13	Placebo	60.00
14	14	Placebo	33.09
15	15	n/a	41.18
16	16	n/a	54.49
17	17	Low	36.66
18	18	High	99.36
19	34	Placebo	48.31
20	35	Low	69.02
21	36	Low	51.71
22	37	High	89.72
23	38	Placebo	87.21
24	39	High	54.41
25	40	Low	79.98
26	41	High	133.16
27	42	Placebo	43.96
28	43	Placebo	98.13
29	44	Placebo	114.35
30	45	Placebo	52.30

Step 1a. Sort & Merge both the files

```
/* Sort both data by PatientID*/  
Proc sort data= L1.Study1 out = L1.Study1_sort;  
by PatientID;  
run;  
Proc print data= L1.Study1_sort;  
Title "Study1_sort";  
run;  
Proc sort data= L1.Study2 out = L1.Study2_sort;  
by PatientID;  
run;  
Proc print data= L1.Study2_sort;  
Title "Study2_sort";  
run;  
/* Merge both data by PatientID- Non_Equi Join*/  
data L1.Final_NonEqui;  
merge L1.Study1_sort L1.Study2_sort;  
by PatientID;  
run;  
proc print data= L1.Final_NonEqui;  
Title "Final_NonEqui";  
run;
```

Study1_sort

Obs	PatientID	Age	State	LOS	Total_Charge
1	1	51	AZ	1	3170
2	2	50	AZ	2	4319
3	3	74	AZ	3	5850
4	4	59	AZ	1	6298
5	5	57	AZ	5	5386
6	6	71	AZ	4	4737
7	7	73	AZ	3	7525
8	8	49	AZ	1	2817
9	9	30	AZ	2	3430
10	11	20	AZ	2	3316
11	12	83	AZ	2	2922
12	13	45	AZ	3	5449
13	14	29	AZ	1	2493
14	15	72	AZ	3	5483
15	16	70	AZ	2	3988
16	17	30	AZ	1	3461
17	18	62	AZ	1	2865
18	19	30	AZ	2	2502
19	20	43	AZ	2	4566
20	21	81	AZ	5	4819
21	22	43	AZ	1	1717
22	23	52	AZ	2	1870
23	24	14	AZ	2	1710
24	25	77	AZ	2	1645
25	26	42	AZ	2	3207
26	27	60	AZ	4	5017
27	29	65	AZ	5	7115
28	30	67	AZ	5	5198
29	31	61	AZ	4	8165
30	32	81	AZ	20	37133

Study2_sort

Obs	PatientID	Group	Test_Score
1	1	Low	46.65
2	2	High	35.36
3	3	Placebo	87.63
4	4	Low	44.60
5	5	Low	59.05
6	6	High	77.41
7	7	Placebo	65.63
8	8	High	49.47
9	9	Low	42.72
10	10	High	83.79
11	11	Placebo	35.36
12	12	Placebo	37.15
13	13	Placebo	60.00
14	14	Placebo	33.09
15	15	n/a	41.18
16	16	n/a	54.49
17	17	Low	36.66
18	18	High	99.36
19	34	Placebo	48.31
20	35	Low	69.02
21	36	Low	51.71
22	37	High	89.72
23	38	Placebo	87.21
24	39	High	54.41
25	40	Low	79.98
26	41	High	133.16
27	42	Placebo	43.96
28	43	Placebo	98.13
29	44	Placebo	114.35
30	45	Placebo	52.30

Final_NonEqui

Obs	PatientID	Age	State	LOS	Total_Charge	Group	Test_Score
1	1	51	AZ	1	3170	Low	46.65
2	2	50	AZ	2	4319	High	35.36
3	3	74	AZ	3	5850	Placebo	87.63
4	4	59	AZ	1	6298	Low	44.60
5	5	57	AZ	5	5386	Low	59.05
6	6	71	AZ	4	4737	High	77.41
7	7	73	AZ	3	7525	Placebo	65.63
8	8	49	AZ	1	2817	High	49.47
9	9	30	AZ	2	3430	Low	42.72
10	10	.	.	.	.	High	83.79
11	11	20	AZ	2	3316	Placebo	35.36
12	12	83	AZ	2	2922	Placebo	37.15
13	13	45	AZ	3	5449	Placebo	60.00
14	14	29	AZ	1	2493	Placebo	33.09
15	15	72	AZ	3	5483	n/a	41.18
16	16	70	AZ	2	3988	n/a	54.49
17	17	30	AZ	1	3461	Low	36.66
18	18	62	AZ	1	2865	High	99.36
19	19	30	AZ	2	2502	.	.
20	20	43	AZ	2	4566	.	.
21	21	81	AZ	5	4819	.	.
22	22	43	AZ	1	1717	.	.
23	23	52	AZ	2	1870	.	.
24	24	14	AZ	2	1710	.	.
25	25	77	AZ	2	1645	.	.
26	26	42	AZ	2	3207	.	.
27	27	60	AZ	4	5017	.	.
28	29	65	AZ	5	7115	.	.
29	30	67	AZ	5	5198	.	.
30	31	61	AZ	4	8165	.	.
31	32	81	AZ	20	37133	.	.
32	33	72	AZ	1	2650	.	.
33	34	18	AZ	1	2893	Placebo	48.31
34	35	51	AZ	4	20598	Low	69.02
35	36	18	AZ	2	3571	Low	51.71

```
246 data L1.Final_NonEqui;  
247 merge L1.Study1_sort L1.Study2_sort;  
248 by PatientID;  
249 run;
```

```
NOTE: There were 3233 observations read from the data set L1.STUDY1_SORT.  
NOTE: There were 3200 observations read from the data set L1.STUDY2_SORT.  
NOTE: The data set L1.FINAL_NONEQUI has 3237 observations and 7 variables.  
NOTE: DATA statement used (Total process time):  
      real time           0.02 seconds  
      cpu time            0.01 seconds
```

```
250 proc print data= L1.Final_NonEqui;  
251 Title "Final_NonEqui";  
252 run;
```

```
NOTE: There were 3237 observations read from the data set L1.FINAL_NONEQUI.  
NOTE: PROCEDURE PRINT used (Total process time):  
      real time           0.36 seconds  
      cpu time            0.36 seconds
```

Step 1b. Remove IDs with missing values

```
/* Remove null values*/  
data L1.Final_Clean;  
    set L1.Final_NonEqui;  
    if cmiss(of _all_) then delete;  
run;  
proc print data= L1.Final_Clean;  
    Title "Final_Clean";  
run;
```

```
286 data L1.Final_Clean;  
287     set L1.Final_NonEqui;  
288     if cmiss(of _all_) then delete;  
289 run;
```

NOTE: There were 3237 observations read from the data set L1.FINAL_NONEQUI.

NOTE: The data set L1.FINAL_CLEAN has 2975 observations and 7 variables.

NOTE: DATA statement used (Total process time):

```
    real time      0.01 seconds  
    cpu time       0.00 seconds
```

```
290 proc print data= L1.Final_Clean;  
291     Title "Final_Clean";  
292 run;
```

NOTE: There were 2975 observations read from the data set L1.FINAL_CLEAN.

NOTE: PROCEDURE PRINT used (Total process time):

```
    real time      0.33 seconds  
    cpu time       0.32 seconds
```

Final_Clean							
Obs	PatientID	Age	State	LOS	Total_Charge	Group	Test_Score
1	1	51	AZ	1	3170	Low	46.65
2	2	50	AZ	2	4319	High	35.36
3	3	74	AZ	3	5850	Placebo	87.63
4	4	59	AZ	1	6298	Low	44.60
5	5	57	AZ	5	5386	Low	59.05
6	6	71	AZ	4	4737	High	77.41
7	7	73	AZ	3	7525	Placebo	65.63
8	8	49	AZ	1	2817	High	49.47
9	9	30	AZ	2	3430	Low	42.72
10	11	20	AZ	2	3316	Placebo	35.36
11	12	83	AZ	2	2922	Placebo	37.15
12	13	45	AZ	3	5449	Placebo	60.00
13	14	29	AZ	1	2493	Placebo	33.09
14	15	72	AZ	3	5483	n/a	41.18
15	16	70	AZ	2	3988	n/a	54.49
16	17	30	AZ	1	3461	Low	36.66
17	18	62	AZ	1	2865	High	99.36
18	34	18	AZ	1	2893	Placebo	48.31
19	35	51	AZ	4	20598	Low	69.02
20	36	18	AZ	2	3571	Low	51.71
21	37	68	AZ	3	14877	High	89.72
22	38	52	AZ	1	3351	Placebo	87.21
23	39	18	AZ	3	3460	High	54.41
24	40	83	AZ	1	995	Low	79.98
25	41	82	AZ	4	7518	High	133.16
26	42	15	AZ	1	2051	Placebo	43.96
27	43	63	AZ	4	6950	Placebo	98.13
28	44	85	AZ	3	5645	Placebo	114.35
29	45	31	AZ	3	5371	Placebo	52.30
30	46	78	AZ	7	6350	n/a	113.83

Step 1c. Remove “n/a” from the variable “Group”

```
/* Remove n/a from Group variable*/  
data L1.Final_Clean2;  
    set L1.Final_Clean;  
    /* Keep only valid treatment groups */  
    if upcase(Group) in ("LOW","HIGH","PLACEBO");  
run;  
proc print data= L1.Final_Clean2;  
    Title "Final_Clean2";  
run;
```

```
293 data L1.Final_Clean2;  
294     set L1.Final_Clean;  
295     /* Keep only valid treatment groups */  
296     if upcase(Group) in ("LOW","HIGH","PLACEBO");  
297 run;
```

```
NOTE: There were 2975 observations read from the data set L1.FINAL_CLEAN.  
NOTE: The data set L1.FINAL_CLEAN2 has 2602 observations and 7 variables.  
NOTE: DATA statement used (Total process time):  
      real time           0.01 seconds  
      cpu time            0.01 seconds
```

```
298 proc print data= L1.Final_Clean2;  
299 Title "Final_Clean2";  
300 run;
```

```
NOTE: There were 2602 observations read from the data set L1.FINAL_CLEAN2.  
NOTE: PROCEDURE PRINT used (Total process time):  
      real time           0.30 seconds  
      cpu time            0.28 seconds
```

Final_Clean2							
Obs	PatientID	Age	State	LOS	Total_Charge	Group	Test_Score
1	1	51	AZ	1	3170	Low	46.65
2	2	50	AZ	2	4319	High	35.36
3	3	74	AZ	3	5850	Placebo	87.63
4	4	59	AZ	1	6298	Low	44.60
5	5	57	AZ	5	5386	Low	59.05
6	6	71	AZ	4	4737	High	77.41
7	7	73	AZ	3	7525	Placebo	65.63
8	8	49	AZ	1	2817	High	49.47
9	9	30	AZ	2	3430	Low	42.72
10	11	20	AZ	2	3316	Placebo	35.36
11	12	83	AZ	2	2922	Placebo	37.15
12	13	45	AZ	3	5449	Placebo	60.00
13	14	29	AZ	1	2493	Placebo	33.09
14	17	30	AZ	1	3461	Low	36.66
15	18	62	AZ	1	2865	High	99.36
16	34	18	AZ	1	2893	Placebo	48.31
17	35	51	AZ	4	20598	Low	69.02
18	36	18	AZ	2	3571	Low	51.71
19	37	68	AZ	3	14877	High	89.72
20	38	52	AZ	1	3351	Placebo	87.21
21	39	18	AZ	3	3460	High	54.41
22	40	83	AZ	1	995	Low	79.98
23	41	82	AZ	4	7518	High	133.16
24	42	15	AZ	1	2051	Placebo	43.96
25	43	63	AZ	4	6950	Placebo	98.13
26	44	85	AZ	3	5645	Placebo	114.35
27	45	31	AZ	3	5371	Placebo	52.30
28	48	78	AZ	3	8658	Low	54.55
29	49	52	AZ	10	13659	High	147.55
30	50	42	AZ	0	3104	Placebo	47.50

Step 2a. Descriptive Stats for Final_Clean2

(to determine if equal sample can be obtained from each category)

```
/* Descriptive statistics on Final_Clean2 to decide about how many from each group.
proc freq data=L1.Final_Clean2;
    tables Group;
run;
```

Group	Frequency	Percent	Cumulative Frequency	Cumulative Percent
High	740	28.44	740	28.44
Low	744	28.59	1484	57.03
Placebo	1118	42.97	2802	100.00

If we sampled proportionally, Placebo would dominate, reducing power for smaller groups.

Sampling method: SRS + Equal Allocation Stratified Sampling to ensure

- Balanced representation of treatment groups,
- Fair comparison-Reduces bias toward larger group (Placebo),
- Makes statistical tests (ANOVA, Kruskal-Wallis) more robust.



Step 2b: Sampling Method= SRS+ Stratified (333 from each group=999 sample size)

```
/* Step 2: Sampling: Method SRS (Random sampling), Sample size 1000 (Don't use seed)*/
proc sort data=L1.Final_Clean2;
    by Group;
run;
proc surveyselect data=L1.Final_Clean2
    method=srs
    sampsize=(333 333 333) /* number from each group */
    out=L1.Final_Sample_SRS_Stratified;
    strata Group;
    id _ALL_;
run;

Proc Print data= L1.Final_Sample_SRS_Stratified;
title "Final_Sample_SRS_Stratified";
run;
```

```
372 proc sort data=L1.Final_Clean2;
373     by Group;
374 run;

NOTE: There were 2602 observations read from the data set L1.FINAL_CLEAN2.
NOTE: The data set L1.FINAL_CLEAN2 has 2602 observations and 7 variables.
NOTE: PROCEDURE SORT used (Total process time):
      real time           0.00 seconds
      cpu time            0.00 seconds

375 proc surveyselect data=L1.Final_Clean2
376     method=srs
377     sampsize=(333 333 333) /* number from each group */
378     out=L1.Final_Sample_SRS_Stratified;
379     strata Group;
380     id _ALL_;
381 run;

NOTE: The data set L1.FINAL_SAMPLE_SRS_STRATIFIED has 999 observations and 9 variables.
NOTE: PROCEDURE SURVEYSELECT used (Total process time):
      real time           0.01 seconds
      cpu time            0.01 seconds
```

Final_Sample_SRS_Stratified

The SURVEYSELECT Procedure

Selection Method	Simple Random Sampling
Strata Variable	Group

Input Data Set	FINAL_CLEAN2
Random Number Seed	202918001
Number of Strata	3
Total Sample Size	999
Output Data Set	FINAL_SAMPLE_SRS_STRATIFIED

Output: Final_Sample_SRS_Stratified

Final_Sample_SRS_Stratified									
Obs	Group	PatientID	Age	State	LOS	Total_Charge	Test_Score	SelectionProb	SamplingWeight
1	High	8	49	AZ	1	2817	49.47	0.45000	2.22222
2	High	18	62	AZ	1	2865	99.38	0.45000	2.22222
3	High	37	66	AZ	3	14877	89.72	0.45000	2.22222
4	High	39	18	AZ	3	3400	54.41	0.45000	2.22222
5	High	41	82	AZ	4	7518	133.16	0.45000	2.22222
6	High	49	52	AZ	10	13059	147.55	0.45000	2.22222
7	High	69	50	AZ	2	5830	66.30	0.45000	2.22222
8	High	71	56	AZ	3	18482	148.71	0.45000	2.22222
9	High	73	59	AZ	1	21194	96.06	0.45000	2.22222
10	High	87	78	AZ	3	9885	37.84	0.45000	2.22222
11	High	97	59	AZ	8	16969	39.03	0.45000	2.22222
12	High	108	70	AZ	5	18919	144.74	0.45000	2.22222
13	High	110	62	AZ	3	27954	185.57	0.45000	2.22222
14	High	112	46	AZ	1	9375	51.73	0.45000	2.22222
15	High	129	88	AZ	2	14300	90.43	0.45000	2.22222
16	High	133	58	AZ	4	17876	66.93	0.45000	2.22222
17	High	143	64	AZ	6	8950	120.21	0.45000	2.22222
18	High	145	56	AZ	0	2721	66.94	0.45000	2.22222
19	High	157	54	AZ	14	60994	334.23	0.45000	2.22222
20	High	177	36	AZ	10	33845	96.57	0.45000	2.22222
21	High	185	55	AZ	3	10578	109.10	0.45000	2.22222
22	High	191	19	AZ	2	12506	41.86	0.45000	2.22222
...	...	...	...	...	...	...	...	...	...

334	Low	1	51	AZ	1	3170	46.65	0.44758	2.23423
335	Low	5	57	AZ	5	5386	59.05	0.44758	2.23423
336	Low	9	30	AZ	2	3430	42.72	0.44758	2.23423
337	Low	36	18	AZ	2	3571	51.71	0.44758	2.23423
338	Low	48	78	AZ	3	8658	54.55	0.44758	2.23423
339	Low	56	80	AZ	5	9575	64.81	0.44758	2.23423
340	Low	68	71	AZ	13	30106	83.43	0.44758	2.23423
341	Low	80	27	AZ	2	3923	26.46	0.44758	2.23423
342	Low	84	28	AZ	6	25435	81.93	0.44758	2.23423
343	Low	88	14	AZ	2	6420	24.64	0.44758	2.23423
344	Low	106	47	AZ	7	16300	112.95	0.44758	2.23423
345	Low	111	81	AZ	1	5910	29.40	0.44758	2.23423
346	Low	131	52	AZ	5	11692	96.51	0.44758	2.23423
347	Low	144	59	AZ	1	9803	74.22	0.44758	2.23423
348	Low	152	71	AZ	5	14287	125.18	0.44758	2.23423
349	Low	155	19	AZ	1	7290	51.59	0.44758	2.23423
350	Low	168	46	AZ	6	17034	60.41	0.44758	2.23423
351	Low	171	80	AZ	2	6622	66.20	0.44758	2.23423
352	Low	172	56	AZ	14	96424	121.09	0.44758	2.23423
353	Low	187	16	AZ	3	8543	47.78	0.44758	2.23423
354	Low	188	28	AZ	2	13063	49.40	0.44758	2.23423
355	Low	203	33	AZ	3	11554	45.28	0.44758	2.23423
356	Low	208	70	AZ	3	15486	132.81	0.44758	2.23423

667	Placebo	11	20	AZ	2	3316	35.36	0.29785	3.35736
668	Placebo	13	45	AZ	3	5449	60.00	0.29785	3.35736
669	Placebo	38	52	AZ	1	3361	87.21	0.29785	3.35736
670	Placebo	45	31	AZ	3	5371	52.30	0.29785	3.35736
671	Placebo	54	24	AZ	1	3087	49.26	0.29785	3.35736
672	Placebo	58	29	AZ	2	4894	92.66	0.29785	3.35736
673	Placebo	59	82	AZ	3	4876	80.42	0.29785	3.35736
674	Placebo	70	48	AZ	2	5806	65.14	0.29785	3.35736
675	Placebo	74	36	AZ	9	30319	42.56	0.29785	3.35736
676	Placebo	82	57	AZ	14	54262	323.16	0.29785	3.35736
677	Placebo	86	30	AZ	3	8713	58.66	0.29785	3.35736
678	Placebo	114	19	AZ	3	11390	50.43	0.29785	3.35736
679	Placebo	130	74	AZ	2	5790	39.09	0.29785	3.35736
680	Placebo	147	77	AZ	5	12042	66.07	0.29785	3.35736
681	Placebo	149	69	AZ	3	16762	86.73	0.29785	3.35736
682	Placebo	154	61	AZ	7	20958	71.77	0.29785	3.35736
683	Placebo	163	45	AZ	6	57437	276.03	0.29785	3.35736
684	Placebo	170	34	AZ	1	2301	27.40	0.29785	3.35736
685	Placebo	181	62	AZ	4	14215	119.66	0.29785	3.35736
686	Placebo	186	36	AZ	1	6937	56.18	0.29785	3.35736
687	Placebo	206	69	AZ	2	2489	46.69	0.29785	3.35736
688	Placebo	213	67	AZ	2	10511	82.23	0.29785	3.35736
689	Placebo	218	62	AZ	6	36941	110.12	0.29785	3.35736
690	Placebo	226	42	AZ	2	12900	100.34	0.29785	3.35736
691	Placebo	229	62	AZ	19	62318	216.47	0.29785	3.35736
692	Placebo	234	57	AZ	17	46765	299.62	0.29785	3.35736
693	Placebo	243	57	AZ	6	38658	271.26	0.29785	3.35736
694	Placebo	244	63	AZ	1	4756	104.13	0.29785	3.35736
695	Placebo	245	54	AZ	4	20023	251.81	0.29785	3.35736
696	Placebo	250	47	AZ	1	2967	38.71	0.29785	3.35736
697	Placebo	261	56	AZ	8	22965	42.00	0.29785	3.35736
698	Placebo	274	72	AZ	2	5182	46.81	0.29785	3.35736
...	...	...	...	...	...	...	...	...	...



Step 2c: Final_Sample_SRS_Stratified: Frequency Procedure

```
/* DESCRIPTIVE STATISTICS */  
/* proc univariate */  
proc freq data=L1.Final_Sample_SRS_Stratified;  
    tables Group;  
run;
```

Final_Sample_SRS_Stratified

The FREQ Procedure

Group	Frequency	Percent	Cumulative Frequency	Cumulative Percent
High	333	33.33	333	33.33
Low	333	33.33	666	66.67
Placebo	333	33.33	999	100.00

Step 3: Run Descriptive Statistics: Mean for “Test_Score”

```
/* Step 3A: Descriptive Statistics */  
PROC MEANS DATA=L1.Final_Sample_SRS_Stratified N MEAN STD MIN MAX MEDIAN;  
    CLASS Group;  
    VAR Test_Score;  
RUN;
```

```
602 PROC MEANS DATA=L1.Final_Sample_SRS_Stratified N MEAN STD MIN MAX MEDIAN;  
603     CLASS Group;  
604     VAR Test_Score;  
605 RUN;
```

NOTE: There were 999 observations read from the data set L1.FINAL_SAMPLE_SRS_STRATIFIED.
NOTE: PROCEDURE MEANS used (Total process time):
 real time 0.03 seconds
 cpu time 0.04 seconds

Final_Sample_SRS_Stratified

The MEANS Procedure

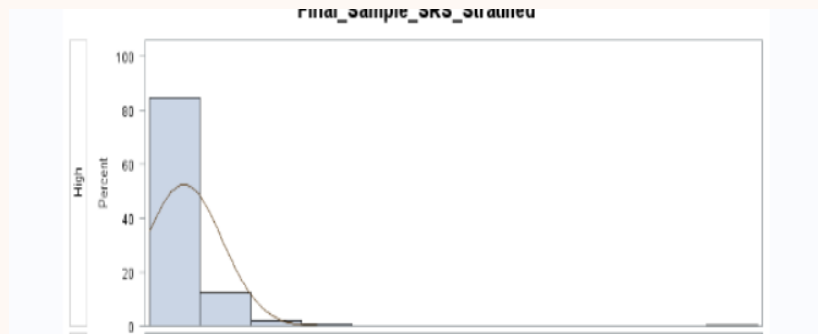
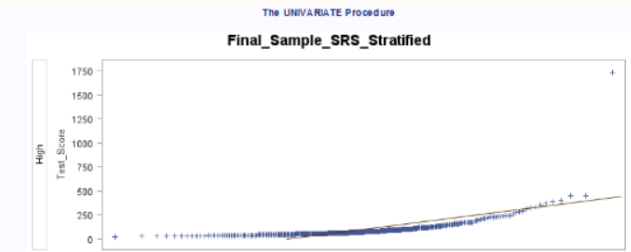
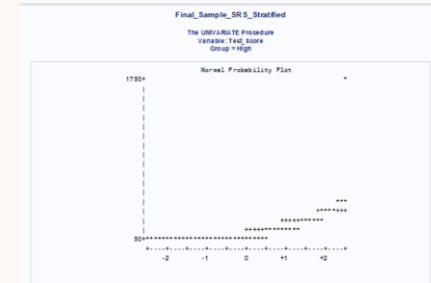
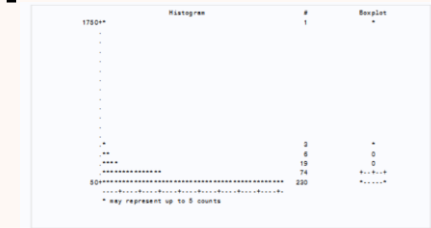
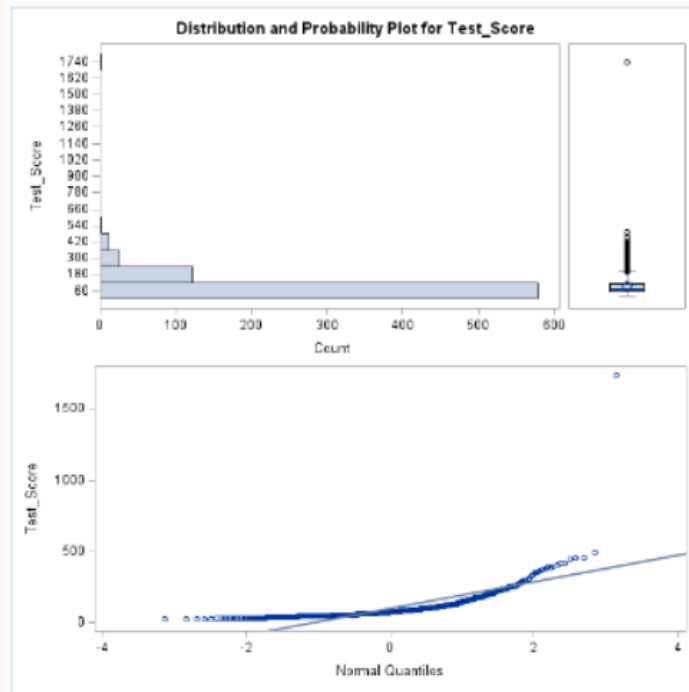
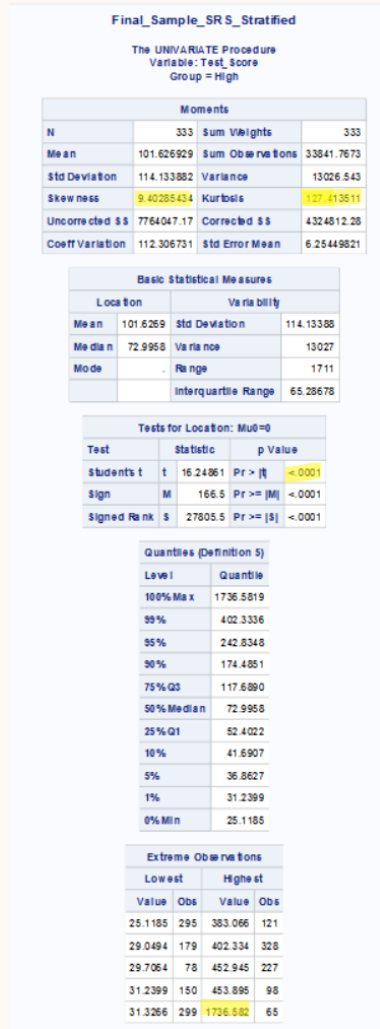
Analysis Variable : Test_Score							
Group	N Obs	N	Mean	Std Dev	Minimum	Maximum	Median
High	333	333	101.6269290	114.1338820	25.1185000	1736.58	72.9958300
Low	333	333	88.3081385	63.7865488	24.6393000	590.2384000	66.2505600
Placebo	333	333	97.0062179	67.0412208	24.6664900	396.0679000	75.9891500

Step 3a: Normality Check using univariate with plots

```
proc univariate data=L1.Final_Sample_SRS_Stratified plots;  
    var Test_Score;  
    class Group;  
    histogram Test_Score / normal;  
run;
```



I. Univariate Procedure Group= "High"



Removed extreme outlier 1736.582 from Group High and re-ran Univariate

```
data L1.Final_Sample_Cleaned;
    set L1.Final_Sample_SRS_Stratified;
    if Group = "High" and Test_Score = 1736.582 then delete;
    title "Final_Sample_Cleaned";
run;

proc univariate data=L1.Final_Sample_Cleaned;
    class Group;
    var Test_Score;
    histogram Test_Score / normal;
    qqplot Test_Score / normal(mu=est sigma=est);
run;
```

Final_Sample_Cleaned

The UNIVARIATE Procedure
Variable: Test_Score
Group = High

Moments			
N	333	Sum Weights	333
Mean	101.626929	Sum Observations	33841.7673
Std Deviation	114.133882	Variance	13026.543
Skewness	9.40285434	Kurtosis	127.413511
Uncorrected SS	7764047.17	Corrected SS	4324812.28
Coeff Variation	112.306731	Std Error Mean	6.25449821

Basic Statistical Measures			
Location		Variability	
Mean	101.6269	Std Deviation	114.13388
Median	72.9958	Variance	13027
Mode	.	Range	1711
		Interquartile Range	65.28678

Tests for Location: Mu0=0			
Test	Statistic	p Value	
Student's t	t 16.24861	Pr > t	<.0001
Sign	M 166.5	Pr >= M	<.0001
Signed Rank	S 27805.5	Pr >= S	<.0001

Quantiles (Definition 5)	
Level	Quantile
100% Max	1736.5819
99%	402.3336
95%	242.8348
90%	174.4851
75% Q3	117.6890
50% Median	72.9958
25% Q1	52.4022
10%	41.6907
5%	36.8627
1%	31.2399
0% Min	25.1185

II. Univariate Procedure Group= “Low,”

Final_Sample_SRS_Stratified

The UNIVARIATE Procedure
Variable: Test_Score
Group = Low

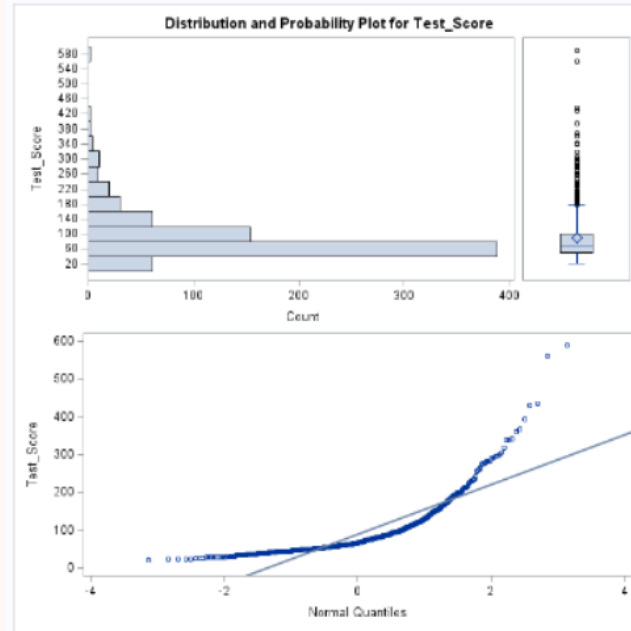
Moments			
N	333	Sum Weights	333
Mean	88.3081385	Sum Observations	29406.8101
Std Deviation	63.7865488	Variance	4068.72381
Skewness	3.28610917	Kurtosis	16.2101195
Uncorrected SS	3947659.31	Corrected SS	1380816.31
Coeff Variation	72.2317896	Std Error Mean	3.49548134

Basic Statistical Measures			
Location		Variability	
Mean	88.30814	Std Deviation	63.78655
Median	66.25056	Variance	4069
Mode	.	Range	565.59910
		Interquartile Range	49.49958

Tests for Location: Mu0=0		
Test	Statistic	p Value
Students t	t 25.26351	Pr > t <.0001
Sign	M 166.5	Pr >= M <.0001
Signed Rank	S 27805.5	Pr >= S <.0001

Quantiles (Definition 5)	
Level	Quantile
100% Max	590.2384
99%	342.4858
95%	203.4333
90%	159.6021
75% Q3	101.2936
50% Median	66.2506
25% Q1	51.7940
10%	42.7179
5%	36.5551
1%	26.4633
0% Min	24.6393

Extreme Observations			
Lowest		Highest	
Value	Obs	Value	Obs
24.6393	343	338.594	425
25.2057	587	342.486	556
25.7857	552	367.688	457
26.4633	341	435.995	484
26.7639	506	590.238	560



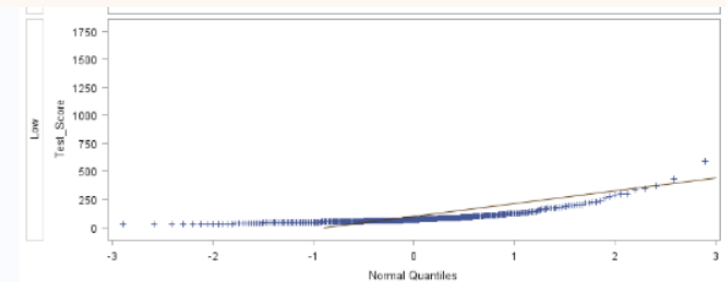
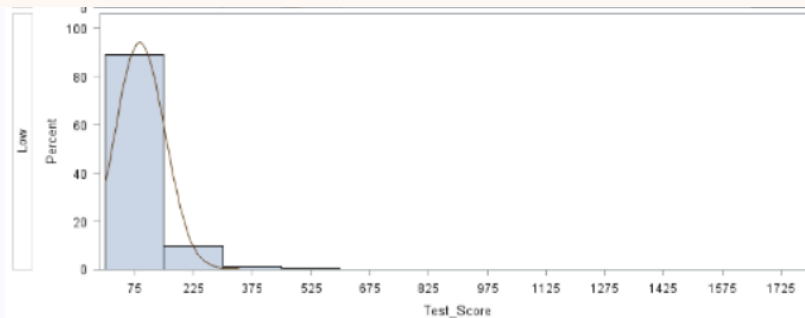
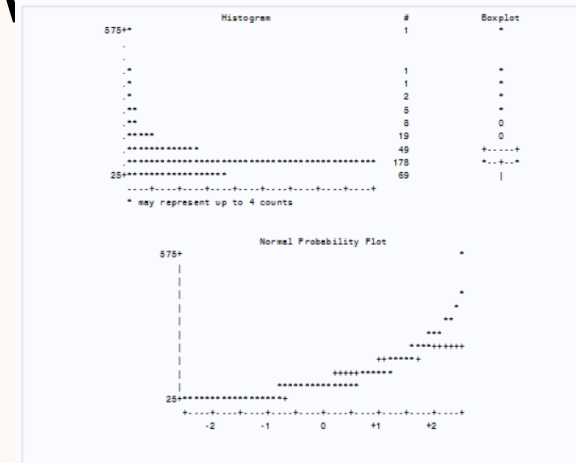
Final_Sample_SRS_Stratified

The UNIVARIATE Procedure
Group = Low
Fitted Normal Distribution for Test_Score

Parameters for Normal Distribution		
Parameter	Symbol	Estimate
Mean	Mu	88.30814
Std Dev	Sigma	63.78655

Goodness-of-Fit Tests for Normal Distribution		
Test	Statistic	p Value
Kolmogorov-Smirnov D	0.1934447	Pr > D <0.010
Cramer-von Mises	W-Sq 4.7093334	Pr > W-Sq <0.005
Anderson-Darling	A-Sq 25.8181018	Pr > A-Sq <0.005

Quantiles for Normal Distribution		
Percent	Observed	Estimated
1.0	26.4633	-60.08156
5.0	36.5551	-16.61140
10.0	42.7179	6.56239
25.0	51.7940	45.28477
50.0	66.2506	88.30814
75.0	101.2936	131.33151
90.0	159.6021	170.05389
95.0	203.4333	193.22767
99.0	342.4858	236.69784



III. Univariate Procedure Group= “Placebo”

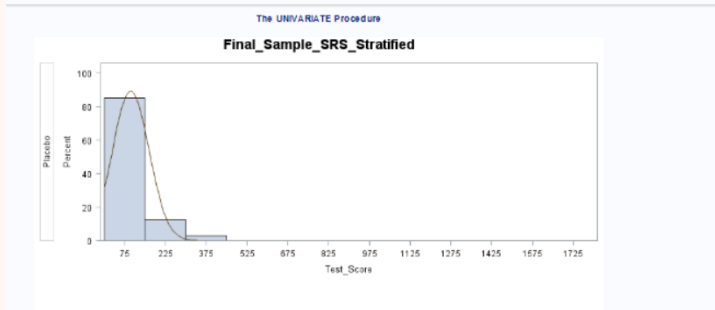
Final_Sample_SRS_Stratified			
The UNIVARIATE Procedure			
Variable: Test_Score			
Group = Placebo			
Moments			
N	333	Sum of Weights	333
Mean	97.0062179	Sum of Observations	32303.0706
Std Deviation	67.0412208	Variance	4494.52528
Skewness	1.9922754	Kurtosis	4.21049533
Uncorrected SS	4625781.1	Corrected SS	1492182.39
Coeff Variation	69.1102305	Std Error Mean	3.67383627

Basic Statistical Measures			
Location		Variability	
Mean	97.00622	Std Deviation	67.04122
Median	75.98915	Variance	4495
Mode	50.03619	Range	371.40141
		Interquartile Range	60.82615

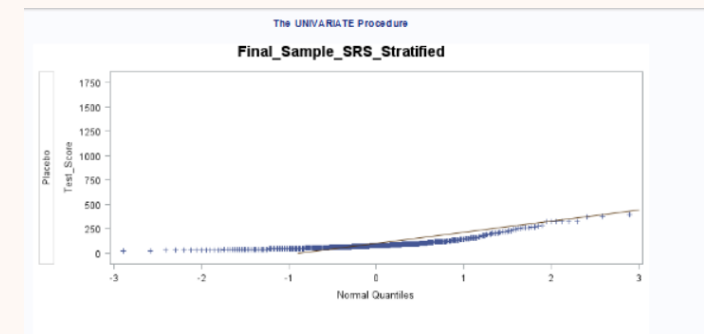
Tests for Location: Mu0=0			
Test	Statistic	p Value	
Student's t	t = 26.40461	Pr > t	<.0001
Sign	M = 166.5	Pr >= M	<.0001
Signed Rank	S = 27805.5	Pr >= S	<.0001

Quantiles (Definition 5)	
Level	Quantile
100% Max	396.0679
99%	326.3809
95%	251.8148
90%	188.2675
75% Q3	113.6762
50% Median	75.9892
25% Q1	52.8501
10%	40.9309
5%	35.5791
1%	27.4014
0% Min	24.6665

Extreme Observations			
Lowest		Highest	
Value	Obs	Value	Obs
24.6665	970	324.630	922
25.9089	973	326.381	700
27.3193	787	368.533	893
27.4014	684	381.460	931
27.6160	875	396.068	888



Final_Sample_SRS_Stratified			
The UNIVARIATE Procedure			
Group = Placebo			
Fitted Normal Distribution for Test_Score			
Parameters for Normal Distribution			
Parameter	Symbol	Estimate	
Mean	Mu	97.00622	
Std Dev	Sigma	67.04122	
Goodness-of-Fit Tests for Normal Distribution			
Test	Statistic	p Value	
Kolmogorov-Smirnov	D = 0.1924797	Pr > D	<0.010
Cramer-von Mises	W-sq = 3.8603734	Pr > W-sq	<0.005
Anderson-Darling	A-sq = 21.8182525	Pr > A-sq	<0.005
Quantiles for Normal Distribution			
Percent	Quantile		
	Observed	Estimated	
1.0	27.4014	-68.9550	
5.0	35.5791	-13.2668	
10.0	40.9309	11.0894	
25.0	52.8501	51.7876	
50.0	75.9892	97.0062	
75.0	113.6762	142.2248	
90.0	188.2675	182.9230	
95.0	251.8148	207.2792	
99.0	326.3809	252.9674	



Summary: Is Test_Score normally distributed?

- No — Test_Score is not normally distributed for High, Low, or Placebo groups even after removing extreme outliers .
- All groups show:
 - Data **violates normality** $p < 0.0001$ for all groups → **Severe non-normality**
- Heavy skewness and outliers make parametric tests unreliable.
- ANOVA assumptions are already broken → No need to proceed with Homogeneity of Variance and PROC ANOVA or PROC GLM
- **Best method:** Kruskal-Wallis for comparing groups.
 - Non-parametric test → does not require normality
 - Kruskal-Wallis is robust for skewed data and outliers.
 - Compares medians across groups
 - Suitable for ordinal or continuous data with skewness/outliers



Appropriate Statistical Test

- We aimed to compare fasting sugar levels across **three independent groups** (Low Drug A, High Drug A, Placebo).
- A **one-way ANOVA** was initially considered, but it requires **normality within each group**.
- **SAS UNIVARIATE** results showed **strong non-normality**:
- These results indicate **violated normality assumptions** and **extreme outliers**.
- Therefore, we used the **Kruskal–Wallis test**, the appropriate **non-parametric alternative** to ANOVA for comparing >2 groups.



Step 4: Kruskal-Wallis Test

```
PROC NPAR1WAY DATA=L1.Final_Sample_SRS_Stratified WILCOXON;  
    CLASS Group;  
    VAR Test_Score;  
RUN;
```

- PROC NPAR1WAY performs non-parametric tests.
- WILCOXON option runs Kruskal-Wallis for >2 groups.
- Output includes Chi-Square statistic, DF, and p-value.

RESULTS: Chi-Square = 3.3251, DF = 2, p-value = 0.1897

Null Hypothesis (H_0): Mean fasting blood sugar levels are equal across High Drug A, Low Drug A, and Placebo.

Alternative (H_1): At least one group differs.

Final_Sample_SRS_Stratified					
The NPAR1WAY Procedure					
Wilcoxon Scores (Rank Sums) for Variable Test_Score Classified by Variable Group					
Group	N	Sum of Scores	Expected Under H0	Std Dev Under H0	Mean Score
High	333	168778.0	166500.0	4299.01150	506.840841
Low	333	158865.0	166500.0	4299.01150	477.072072
Placebo	333	171857.0	166500.0	4299.01150	516.087087
Average scores were used for ties.					
Kruskal-Wallis Test					
Chi-Square	DF	Pr > ChiSq			
3.3251	2	0.1897			

Results of Kruskal-Wallis Test

Interpretation

- **$p = 0.1897 > 0.05$** → Fail to reject H_0 .
- There is **no statistically significant difference** in mean fasting blood sugar levels among the three groups after 10 weeks.
- Even though the rank sums differ slightly (Placebo > High > Low), these differences are **not significant**.



Conclusion

Research Question: Is there a significant difference in mean fasting sugar levels (Test_Score) among the three treatment groups: Low Drug A, High Drug A, and Placebo after 10 weeks?

- Drug A (in both high and low doses) did **not show a significant effect** compared to Placebo in reducing fasting blood sugar.
- The data is highly non-normal with wide spreads of test score values.
- Non-parametric results showed no evidence of a treatment effect across the three groups.
- Findings suggest that Drug A is not effective in lowering fasting glucose under the conditions tested.



Acknowledgement

Thank you
Professor Ridhi Vyas
for
Guidance and Support

